

PRINT Name: _____

Test Version: 121 One-Answer Multiple Choice 25 Questions – 10 of 20%

☞ Fill in the bubbles for your Name and Answers with pencil only, no pen.

☞ Manage your time. Answer questions you know, first. One Answer per question.

1. What is a **functor** in Prolog?
 - a. Any one of the arguments of a structure.
 - b. All of these answers.
 - c. None of these answers.
 - d. The name (an atom) at the beginning of a structure, together with its arity.
 - e. The head of a clause that includes the `:-` symbol.
2. What is declarative programming?
 - a. All of these answers.
 - b. A programming paradigm where a solution is achieved when message passing between objects stabilizes.
 - c. A programming paradigm where the programmer describes the solution rather than the steps required to produce the solution.
 - d. A programming paradigm where the programmer declares ordered operations for building a solution.
 - e. None of these answers.
3. What would be the result of the following query in Prolog?
 $X = [x], \text{ \texttt{\textbackslash+ } } X = [x, y, z] .$
 - a. $X = [x]$
 - b. Variable instantiation error.
 - c. None of these answers.
 - d. false
 - e. $X = [x, y, z]$
4. Which of the following queries will turn on tracing during a Prolog session?
 - a. None of these answers.
 - b. `notrace`.
 - c. `trace`.
 - d. `guitracer`.
 - e. `noguitracer`.
5. What is a **structure** generally in Prolog?
 - a. All of these answers.
 - b. None of these answers.
 - c. Only the body of a clause that includes the `:-` symbol.
 - d. Only any clause that includes the `:-` symbol.
 - e. An object that has several components, namely, a functor and arguments.

6. In Prolog, what is the difference between `[a]` and `[h|[]]`?
 - a. `[h|[]]` is a syntax error and `[a]` is a list.
 - b. There is no difference, as they both mean `[a]`.
 - c. `[a]` has one element and `[h|[]]` has two elements.
 - d. Each is a one-element list with a different element.
 - e. None of these answers.
7. What is the meaning of `Q :- P.` in a Prolog program?
 - a. If Q then P.
 - b. Q is true if P is true.
 - c. None of these answers.
 - d. Q implies P.
 - e. All of these answers.
8. What is the difference between **matching** and **unification**?
 - a. None of these answers.
 - b. **unification** results in variable bindings, whereas **matching** results in variable values.
 - c. All of these answers.
 - d. There is no difference.
 - e. **matching** a variable with a structure involves checking to ensure the variable does not occur within the structure.
9. Which of the following is a variable in Prolog?
 - a. x
 - b. `var x`
 - c. 'X'
 - d. A
 - e. All of these answers.
10. Which of the following terms is a Prolog atom?
 - a. `var`
 - b. None of these answers.
 - c. 3
 - d. `_`
 - e. B
11. Which of the following statements about the cut (!) goal in Prolog is true?
 - a. It can increase efficiency by reducing unnecessary backtracking.
 - b. None of these answers.
 - c. All of these answers.
 - d. In some situations it changes the logical meaning of a program.
 - e. We avoid it in this course if possible because we try for purely logical Prolog
12. What is the result of the following Prolog query? `X is 7 + 2.`
 - a. $X = '7 + 2'$
 - b. None of these answers.
 - c. $X = '9'$
 - d. $X = 9$
 - e. $X = 7 + 2$

13. What is meant by **arity** in Prolog?
- The number of answers that a query can produce.
 - All of these answers.
 - None of these answers.
 - The number of clauses that define a single predicate.
 - An integer associated with a functor or predicate, indicating its number of arguments.
14. Which of the following statements about negation in Prolog is true, given some goal **G**?
- $\backslash + \text{ G}$ is safest if any variables within **G** are already instantiated.
 - $\backslash + \text{ G}$ will always fail if **G** contains uninstantiated variables.
 - $\backslash + \text{ G}$ will always succeed if **G** contains uninstantiated variables.
 - None of these answers.
 - $\backslash + \text{ G}$ is safest if none of the variables within **G** are instantiated.
15. What is the result of the following Prolog query? $\text{X} = 1 + 5$.
- None of these answers.
 - $\text{X} = 6$
 - $\text{X} = '1 + 5'$
 - $\text{X} = '6'$
 - $\text{X} = 1 + 5$
16. What is the result of the following Prolog query? $\text{X is Y} + 4, \text{Y} = 4$.
- $\text{X} = 4, \text{Y} = 4$
 - $\text{X} = 8, \text{Y} = 4$
 - None of these answers.
 - $\text{X} = 8$
 - Variable instantiation error.
17. What would be the result of the following Prolog query?
- $$\text{a} * \text{b} = *(\text{a}, \text{b}) .$$
- Syntax error
 - $\text{a} = \text{a}, \text{b} = \text{b}$
 - None of these answers.
 - true
 - false
18. What is the **arity** of a Prolog predicate?
- The number of arguments the predicate takes.
 - None of these answers.
 - The number of clauses used to define the predicate.
 - The number of variables used to define the predicate.
 - The number of answers the predicate can produce when queried.
19. Which of the following statements is true about the anonymous variable in Prolog?
- When it occurs more than once in a clause, it can match different things.
 - All of these answers.
 - None of these answers.
 - It is written as a single underscore, `_`.
 - When a variable appears only once in a clause, it can be replaced by the anonymous variable to avoid the singleton variable warning.

20. What would be the result of the following query in Prolog?
- $$\backslash + \text{ X} = [\text{x}, \text{y}, \text{z}] .$$
- $\text{X} = []$
 - None of these answers.
 - false
 - Variable instantiation error.
 - true
21. What is the meaning of $\text{P} :- \text{Q}, \text{R} .$ in a Prolog program?
- From Q and R follows P.
 - If Q and R, then P.
 - P is true if Q and R are true.
 - None of these answers.
 - All of these answers.
22. What is the Closed World Assumption (CWA) in Prolog?
- The assumption that if something cannot be proven to be true, then it is false.
 - None of these answers.
 - All of these answers.
 - The assumption that no new clauses will be added to a Prolog program while it is running.
 - The assumption that the results of a specific query cannot change during a Prolog session.
23. How would we write **if X is a student, then X is a person** as a Prolog rule?
- $\text{student}(\text{X}) :- \text{person}(\text{X}) .$
 - $\text{isa}(\text{X}, \text{student}) :- \text{isa}(\text{X}, \text{person}) .$
 - $\text{student}(\text{X}) :- \text{X}, \text{person}(\text{X}) .$
 - $\text{person}(\text{X}) :- \text{student}(\text{X}) .$
 - $\text{X isa student} :- \text{X isa person} .$
24. Which statement about the **consult** predicate in Prolog is true?
- [file]** . is an abbreviation for **consult (file)** .
 - All of these answers.
 - It reads Prolog source code from a file.
 - [file]** . is an abbreviation for **['file.pl']** .
 - When consulting, the special file **user** indicates the Prolog source code should be read from the user at the terminal.
25. What variable bindings would result from the following Prolog Query?
- $$[\text{H}|\text{T}] = [\text{a}, \text{b}, \text{c}] .$$
- $\text{H} = [\text{a}], \text{T} = [\text{b}, \text{c}]$
 - None of these answers.
 - $\text{H} = [\text{a}, \text{b}, \text{c}], \text{T} = []$
 - $\text{H} = \text{a}, \text{T} = [\text{b}, \text{c}]$